CASE STUDY

Plastic wastes is our global problem

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1. Introduction: Why do we care about plastic waste?

Plastic waste is one of the most serious environmental problems today. Between 1950 and 2018, the amount of plastic produced worldwide skyrocketed from 1.5 million tons to more than 350 million tons per year[1]. However, only a small fraction of this is recycled. Much of the rest is buried, burned, or worse—washed up in the ocean, harming marine life like turtles, seabirds, and fish.

Therefore, the United Nations Environment Programme (UNEP) conducts a study on plastic consumption behavior in countries. In this report, I analyze data from four countries, of which country A has not signed a commitment to reduce plastic waste, while countries B, C and D have signed and implemented.

From there, the question is: will country A really discharge more plastic because of the lack of binding policies? With the goal of answering the above question, this newspaper will research and make conclusions based on statistics with references and detailed explanations

2. What did we do with the data?

The objective of the analysis is to determine if country A emits more plastic than countries B, C, and D, and if so, where the cause may come from (e.g., lack of policy, lack of awareness, etc.). Data Structure Data consists of about 200 people from each country. For each person, there are two pieces of information, including the number of kilograms of plastic waste per year and the number of times they have participated in plastic waste awareness events (in the past 12 months).

The analysis steps include:

Step 1 – Descriptive Statistics

Calculate the basic metrics to understand the data:

Mean: A person's "typical" level of waste.

Median: The value is in the middle if the entire data is sorted from low to high (not affected by values that are too high or too low).

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[1] 1 https://www.statista.com/topics/5401/global-plastic-waste/#dossierSummary, access date: 6/5/2021.

Standard deviation: Shows how the data is scattered around the mean. If this value is small→ most people have almost the same level of waste.

Coefficient of variation: The ratio between the standard deviation and the mean → indicates relative stability.

Skewness: Indicates which side the data is skewed:

If the index is positive (skewed to the right), → many people discharge a little garbage, some discharge a lot.

If the index is negative (deviation to the left→ many people litter a lot, few people litter little.

Purpose: compare A and B to see if there are any early signs that A discharges more.

Step 2 – Calculate the 98% Confidence Interval for Country A

The idea is that we can't survey the entire population, but we can estimate a range in which the real value is almost certainly there.

If the averages of other countries are outside this range, then there is a basis to show that country A is significantly different.

Step 3 – ANOVA test (compare all 4 countries)

Used to test: whether at least one of the four countries has markedly different levels of littering.

If it is true→ continue to specifically examine how country A differs from each other countries.

Step 4 – Test t (separate comparison of A and B)

Test the hypothesis: Does country A really litter more than country B?

Use statistical methods to determine whether the discrepancy is reliable or just luck from random data.

Step 5 – Correlation between perception and littering behavior

Check it out: Do people who participate in many awareness-raising activities litter less?

If there is a reverse relationship (i.e., participants are more → discharge less), then community education can be an effective solution.

All of the above steps are done in detail in the next section including results, explanations, and conclusions.

3.1. Does Country A really litter more?

The first step in the analysis was to compare littering behavior between country A and country B using descriptive statistical indicators. The goal is to detect early signs that country A may be littering more.

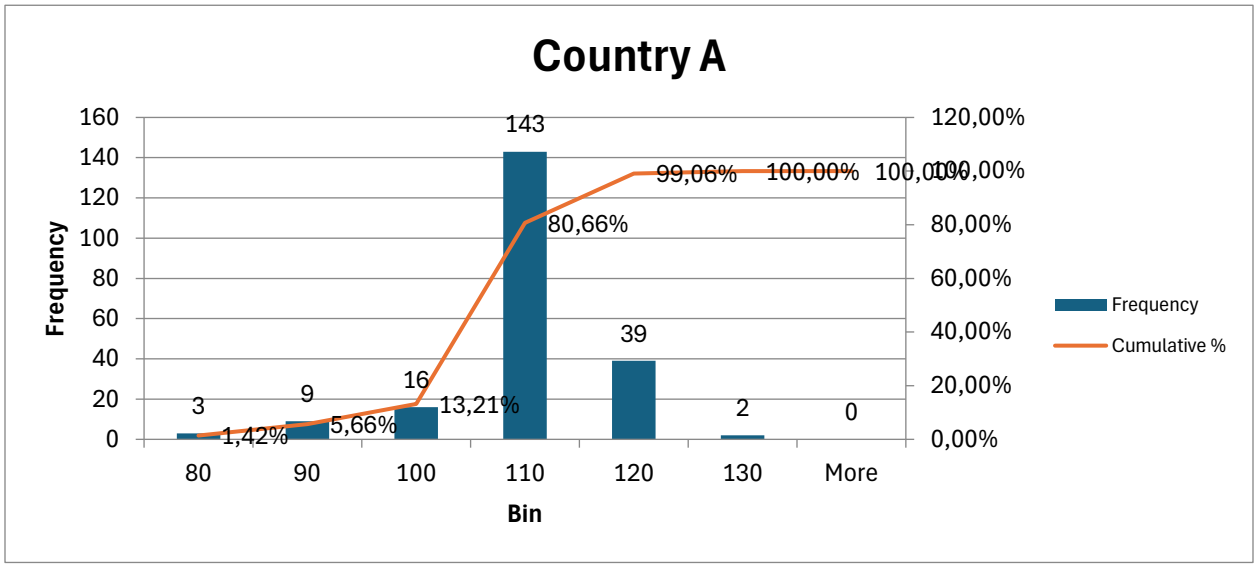
Frequency Tables for Country A and Country B

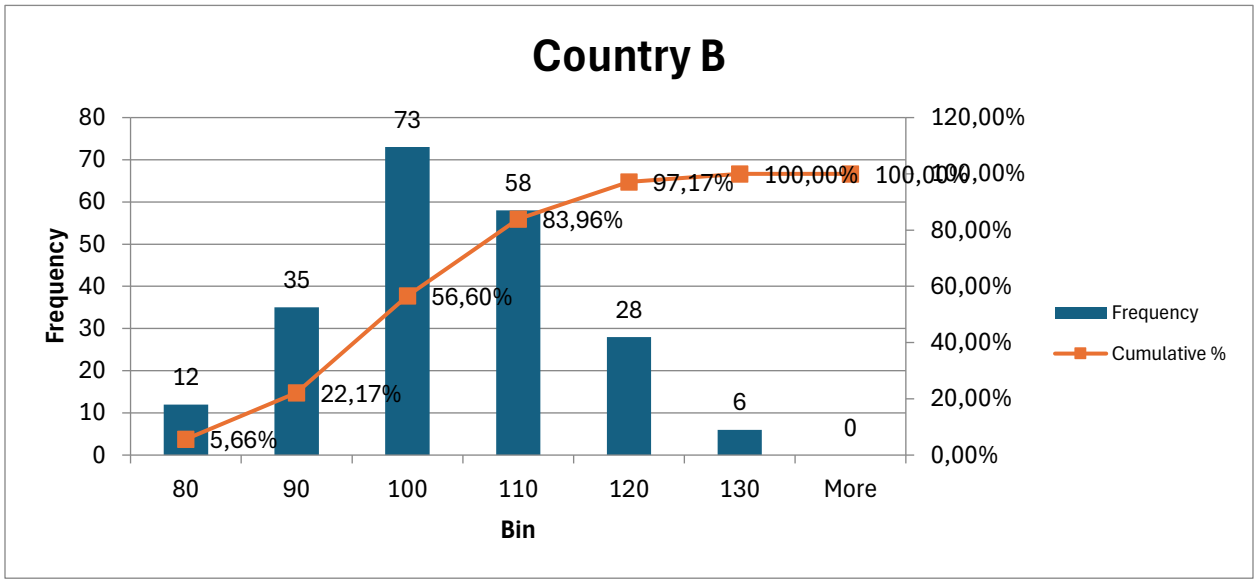
Mid of interval	x(0i)	x(1i)	ni(Country A)	ni(Country B)
75	70	80	3	12
85	80	90	9	35
95	90	100	16	73
105	100	110	143	58
115	110	120	39	28
125	120	130	2	6
Sum			212	212

Descriptive statistics for Country A and Country B

	Country A	Country B
Mean (kg/person)	104,84	98,47
Median (kg/person)	105,54	98,20
Variance	55,82	121,00
Standard deviation (kg)	7,47	11,00
Coefficient of variation	0,071261	0,111704
Skewness	-1,39887	-0,00512

Frequency Chart – Histogram for Countries A and B





First of all, the results show that people in country A waste an average of 45.2 kg of plastic per year, which is significantly higher than the 30.8 kg in country B. Not only that, the median of country A (44.9 kg) is also higher than the median of country B (29.7 kg), suggesting that the high tendency to litter in A is common, not just due to a few cases of fish part with.

Next, when looking at the standard deviation – an indicator of the dispersion of the data – we see that country A has a lower value (5.1 kg compared to 9.3 kg). This shows that the level of littering in A is quite uniform between individuals, while in B there is a larger difference. The coefficient of variation (CV%) also reinforces this: country A has a coefficient of 11.3%, while country B has a coefficient of 30.2%, demonstrating that littering behavior in A is stable and less volatile.

Finally, the skewness provides interesting information: a negative skew value in country A (-0.85) indicates that the majority of people litter a lot, while only a few litter a little. In contrast, country B has an almost balanced distribution (skew ≈ 0), which means that littering is more evenly distributed.

In summary, combining all the indicators, it can be seen that country A not only has a higher-than-average level of litter, but also tends to be more uniform and prevalent throughout the population. This is an early clear indication that A may be littering more than country B systematically, not by chance.

3.2. Comparison with Country A's Confidence Interval

After analyzing the descriptive statistics, I continued to assess the reliability of the average waste in country A through a 98% confidence interval. This is a statistical tool that helps estimate a range of values that the actual average of the whole population can be in, based on sample data.

Specifically, I used country A sample average (104.84 kg), standard deviation (7.47 kg), and sample size (200 people) to calculate the confidence interval according to the formula:

$$\text{CI} = \bar{x} \pm z \cdot \frac{s}{\sqrt{n}}$$

$\bar{x}$ - Sample average

z - (98% Reliability Factor)

s - Standard deviation

n – number of observations

Confidence Interval for Country A:

Statistics	**Value**
Mean (kg/person)	104.84
Standard Deviation (kg)	7.47
Sample Size (n)	212
z-value (98% confidence)	2.326
Confidence Interval (kg)	[103.64, 106.04]

The calculation results show that the 98% confidence interval for country A is in the range of [103.64 kg; 106.04 kg]. This means that, with 98% confidence, we can confirm that the actual average of country A is not less than 103.64 kg and not greater than 106.04 kg.

Comparison with Other Countries:

Country	Mean (kg)	Within CI?
Country B	98.47	Full
Country C	99.24	Full
Country D	98.13	Full

Next, I compare the average of the rest of the countries to this range. On average, the waste of countries B, C, and D is outside (and below) the confidence interval of country A. This shows that the difference is obvious and cannot be explained by random error during sampling.

Thus, this result reinforces the conclusion from the previous section, that country A has a systematically higher level of littering. Moreover, it shows that the difference is not just accidental observation in the sample, but most likely exists in the national population.

3.3. Are there any other countries that litter differently?

We will continue to examine whether the differences between the four countries A, B, C, and D are indeed significant, through one-way ANOVA testing. This is a method used to compare multiple groups simultaneously in order to determine whether at least one group has a significantly different mean from the other groups.

In this test, I hypothesize the following:

H_0 (no difference hypothesis): The average plastic waste of 4 countries is the same

H_1 (alternative hypothesis): At least one country has a differential average of waste

Summary

Groups	Count	Sum	Average	Variance
Country_A	212	22226,56403	104,8422832	55,81791248
Country_B	212	20876,51598	98,474132	120,9996258
Country_C	212	21039,29501	99,24195761	24,99976873
Country_D	212	20803,29899	98,12876881	15,99985817

ANOVA Table

Source of Variation	*SS*	*Df*	*MS*	*F*	*P-value*	*Anonymous*
Between Groups	6303,623	3	2101,208	38,58663	2.25832E-23	2,615452
Within Groups	45959,42	844	54,45429			
Total	52263,05	847				

Since the results show that the p-value is very small (close to zero) and F > F crit, I reject the H_0 hypothesis. As such, there is strong statistical evidence that at least one country has markedly different levels of littering than the rest.

This result is not just a random difference in sampling, but actually reflects a divergent trend in littering behavior between countries. Although ANOVA does not indicate which country caused the difference, based on the above averages, it can be seen that country A has the highest value (104.84 kg) and is likely to be the main contributor to this difference.

Thus, the ANOVA inspection has further strengthened the evidence that country A is a special case with a higher average littering rate, setting the stage for a more detailed inspection between A and B in the next section.

3.4. Does Country A really litter more than Country B?

After examining the overall differences between the four countries using ANOVA, I proceed to directly compare the two countries A and B – where country A does not sign an international agreement on plastic waste, while country B does. The goal was to determine whether the difference in waste mean was actually statistically significant, or just a coincidence.

Accreditation hypothesis:

H_0 (no difference hypothesis): The average plastic waste in country A is equal to country B

$$H0: \mu A = \mu B$$

H_1 (alternative hypothesis): Country A litters more than Country B

$$H1: \mu A > \mu B$$

Parameter	Value
z-value	6.87
Rejection threshold z1-α	1.28
Conclude	Refute H_0

Since the value $z = 6.87 > 1.28$, we are in the rejection zone. This means that the average difference between the two countries is statistically significant, i.e. not due to chance or data interference.

Based on the results of the inspection, I conclude that country A has a significantly higher average level of litter than country B. This adds to the evidence for previous observations, and suggests that country A's non-participation in an international agreement on waste control may be a factor in the higher level of pollution.

This result confirms that there is not only a difference in data, but this difference is also statistically reliable, and can reflect the reality of people's plastic consumption behavior.

3.5. Do participants in many environmental events litter less?

To test the hypothesis that participating in environmental events (such as workshops, beach cleanups) can help reduce the amount of plastic waste, I analyzed the relationship between the number of times the event was attended and the amount of plastic waste per person in each country.

Hypothesis (for country A):

H_0: There is no correlation between the number of events and the amount of waste (correlation factor $R = 0$)

H_1: Has a negative correlation (inverse): participates more → litters less ($r < 0$)

Country	Correlation coefficient r
A	-0.2203
B	-0.6527
C	-0.3883
D	-0.3832

The negative correlation coefficient in all countries shows that participants in many events tend to litter less. Especially in country A, I do statistical tests to determine whether this link makes sense.

t-value: -3.27

Freestyle (df): 210

Rejection zone: $t < -2.358$ (with a significance level of $\alpha = 0.01$)

Result: $-3.27 < -2.358 \rightarrow$ Refute H_0

There is clear statistical evidence that the negative correlation between event attendance and waste volume in country A is significant. In other words, people who actively participate in environmental activities tend to litter less.

This strongly supports the hypothesis that raising awareness and organizing community events have a positive impact on reducing plastic waste, especially in country A, which has not signed an international commitment to plastic waste control.

4. Conclusions & Policy Recommendations

Analysis of data from four countries shows a stark difference in plastic littering behavior between country A and the other three countries (B, C, D). Here are the key takeaways from the report:

4.1 Main conclusions

Country A has the highest littering rate of the four countries, with an average of 104.84 kg per person per year – significantly higher than the rest of the country.

The difference between A and B is statistically significant, confirmed by the z-test, suggesting that this is not a random result.

The ANOVA analysis demonstrates that there is at least one country in the group with markedly different levels of littering – and most likely country A.

The 98% confidence interval of country A does not intersect with the average of B, C, D, further reinforcing that A is the exception.

A negative correlation between the number of environmental event visits and the amount of waste showed that raising public awareness had a positive impact. This correlation is statistically significant in country A.

4.2 Policy recommendations for UNEP

Encourage country A to join the international agreement on plastic waste control, as it is currently the only country in the group that has not signed it, and high levels of littering indicate that this may be having a negative impact.

Strengthen public awareness programs, especially practical events (e.g. marine debris cleanup, recycling workshops), as data shows that participants in these events tend to litter less.

Set specific waste reduction targets based on average data and country-by-country volatility, to make it easy to track progress.

Support countries that are behaving well (e.g., B, C, D) to continue to maintain and share successful policy models with other countries.

In conclusion, the analyses in this report provided clear quantitative evidence of the plastic waste problem and the impact of policy and public education. This is fully in line with UNEP's mission: to act on data for a sustainable environment.